CHANDU PRASAD BHAIRAPU



AI Engineer

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ABOUT ME

Master's graduate in Artificial Intelligence with 3 years of professional experience in AI engineering and software development. Specialized in Python backend services, cloud deployment on AWS and Azure with Docker and CI/CD, and ML data pipelines for generative AI and agentic systems. Most recently sole developer of an agentic knowledge-extraction system for the German energy sector, backed by a strong foundation in machine learning, data analysis and Python development across enterprise and start-up environments.

EXPERIENCE

Associate Generative AI Developer - Agentic Indexing Pipeline Engineer for German Energy Sector

Kniit GmbH

November 2025 – present Muldestausee, Germany

- Sole developer owning the full knowledge-extraction data pipeline end-to-end – data ingestion, chunking, prompt engineering, and indexing (agentic, deterministic, and monolithic strategies) – reporting directly to the CTO and collaborating with SAP domain experts across the German energy sector.
- Developed Expert Interface, a graph curation web app (React frontend, Python microservices backend) deployed on Azure, enabling domain experts to validate and approve knowledge-graph changes.
- Designed and implemented a Model Context Protocol (MCP) query service exposing the knowledge graph for millisecond-latency retrieval, integrated directly into Claude Code for agentic development workflows.
- Engineered a deterministic (code-based) extraction pipeline for a second, high-volume source stream, cutting a $200,000 projected LLM budget to near-zero recurring cost while pushing extraction accuracy toward 100%.
- Designed and optimized multi-agent extraction algorithms (LangGraph: entity-extraction, relationship-analysis, consolidation, and reflection agents running in parallel) for automated annotation of complex regulatory and technical documents, driving precision/recall from 33%/45% to 93%/90% via an SAP-system-derived ground truth.
- Delivered as an AI-augmented engineer – leveraging Claude Code, custom skills, and agentic coding workflows for rapid, high-quality development grounded in clean code principles.
- **Tech stack: Python, Docker, Kubernetes, Azure, FastAPI, microservices, MCP, LangGraph, LangChain, LiteLLM, React, Neo4j, MongoDB, Postgres, GraphRAG, RAG, Vector Search**

GenAI Developer (AI Consulting Intern) - Development and Deployment of an Agentic AI Workflow

Covestro Deutschland AG

December 2024 – April 2025 Leverkusen, Germany

- Designed and deployed an agentic AI application using LangGraph to optimize decision-making workflows for vector database queries, general inference, and summarization tasks.
- Developed a React-based front-end and a Python backend with FastAPI, leveraging Postman for API testing.
- Built and containerized the entire stack with Docker and deployed it to AWS using CDK, adhering to clean code practices. Version-controlled via GitLab, with CI/CD integration.
- **Tech stack: Python(Backend), AWS, Docker, CDK (infrastructure-as-code), GitLab CI/CD, EC2, Lambda, FastAPI, React(Frontend), LangGraph, LangChain, Postman**

GenAI Developer (Master's Thesis) - Evaluating the Performance of RAG Systems

Covestro Deutschland AG

May 2024 – November 2024 Leverkusen, Germany

- Automated RAG's Generation and Retrieval evaluation with RAGAS and PytrecEval, comparing model answers to chemist-provided ground truth.
- Built AWS cloud infrastructure with Lambda, SQS, and DynamoDB for PrismaChat RAG evaluation using CDK.
- Benchmarked multilingual embedding models on over 1M data chunks of Covestro's internal chemical data.
- Designed RAG pipelines, focusing on chunking strategies, vector databases, indexing, and fine-tuning models for optimized retrieval.

- **Tech stack: Python, AWS (S3, EC2, VPC, Lambda, SQS, DynamoDB, CDK), Docker, PyTorch, TensorFlow, Scikit-Learn, LangChain, Hugging Face Transformers, RAGAS, PytrecEval, Azure AI Search, Streamlit, Vector Databases**

Data Scientist (Intern) – ML-Based Prediction of Prepolymer Synthesis Outcomes

Covestro Deutschland AG

November 2023 – April 2024 | Leverkusen, Germany

- Built and compared Random Forest, XGBoost, and LSTM models to predict prepolymer synthesis outcomes (e.g. viscosity, reaction progression) from time-series process data, evaluating tree-based vs. deep learning approaches for this tabular/temporal hybrid problem.
- Engineered lag features and rolling statistical windows from raw sensor time-series (temperature, reaction time, process variables) to feed the tree-based models, while feeding raw sequences directly into the LSTM.
- Benchmarked model performance against traditional empirical curve-fitting methods, improving early identification of batches likely to deviate from target specification.
- Deployed an interactive Streamlit dashboard exposing model predictions to chemists in real time, letting them simulate parameter changes and see predicted outcomes before running physical batches.
- **Tech stack: Python, Scikit-Learn, XGBoost, TensorFlow/Keras, Pandas, NumPy, SciPy, Matplotlib, Streamlit**

Machine Learning Engineer - Electrical Vehicle Load Forecasting using Time Series Analysis

Cognizant Technology Solutions

September 2021 – March 2022 | Hyderabad, India

- Developed ARIMA, SARIMA, XG-Boost regressor, FB Prophet, LSTM models for time series analysis for EVs.
- Developed LSTM, achieved 96% test accuracy and strong performance on unseen data.
- **Tech stack: Python, Scikit-Learn, TensorFlow**

PROJECTS

Project: Robot Episode Ingestion Pipeline | Python | PyAV | FFmpeg | AWS S3 | Terraform | Docker | GitHub Actions

- Python service that ingests multi-modal recording episodes (video, depth, joint states), validates data quality (dropped frames, timestamp drift), compresses video with H.264/AV1 via PyAV, and uploads training-ready episodes to S3 with a manifest, with CLI tools for operators. S3 and IAM provisioned with Terraform, packaged with Docker, tested in GitHub Actions.

TECHNICAL SKILLSET

- **Programming:** Python
- **Backend & APIs:** FastAPI, Python microservices, REST APIs, Postman, React(beginner)
- **Cloud & DevOps:** AWS - S3, EC2, Lambda, SQS, Load balancer, DynamoDB, VPC, CDK (infrastructure-as-code); Docker, Kubernetes, CI/CD (GitLab), Azure, Git
- **Data & ML pipelines:** Machine Learning (Scikit-Learn), Deep Learning (PyTorch, TensorFlow), Pandas, NumPy, SQL, Postgres, MongoDB, Vector Databases(Qdrant, Pinecone, Azure AI Search)
- **GenAI & Agentic Workflow:** LangGraph, LangChain, MCP, Hugging Face transformers, NLTK, RAG
- **Methodologies:** Agile
- **Languages:** English (Business Fluent), German (B1), Hindi (Fluent), Telugu (Native)

EDUCATION

Master of Science in Artificial Intelligence — March 2022 – April 2025
Technische Hochschule Würzburg-Schweinfurt, Würzburg, Germany

Minor in Computer Science & Engineering — July 2019 – July 2021
G Pulla Reddy Engineering College, Kurnool, India

Bachelor of Technology in Mechanical Engineering — August 2017 – July 2021
G Pulla Reddy Engineering College, Kurnool, India

PUBLICATIONS & CERTIFICATIONS

- **[1]** Navigating the Future: An approach of Autonomous Indoor Vehicles
- Udemy - LLMs Mastery : Transformers & Generative AI